

Consider a disk queue with requests for I/O to blocks on cylinders 98, 183, 41, 122, 14, 124, 65, 67. The SSTF scheduling algorithm is used. The head is initially at cylinder number 53. The cylinders are numbered from 0 to 199. Write a Program to calculate the total head movement (in number of cylinders) incurred while servicing these requests.

Code :

```
#include <stdio.h>
#include <stdlib.h>

int main()
{
    int req[100], visited[100];
    int n, head, i, j;
    int total = 0;

    printf("Enter number of requests: ");
    scanf("%d",&n);

    printf("Enter request sequence:\n");
    for(i=0;i<n;i++)
    {
        scanf("%d",&req[i]);
        visited[i] = 0;
    }

    printf("Enter initial head position: ");
    scanf("%d",&head);

    for(i=0;i<n;i++)
    {

```

```
int min = 999, index = -1;

for(j=0;j<n;j++)
{
    if(!visited[j])
    {
        int dist = abs(head - req[j]);
        if(dist < min)
        {
            min = dist;
            index = j;
        }
    }
}

total += min;
head = req[index];
visited[index] = 1;
}

printf("Total Head Movement = %d", total);

return 0;
```

Output:

```
Enter number of requests: 8
Enter request sequence:
98 183 41 122 14 124 66 67
Enter initial head position: 53
Total Head Movement = 323

=== Code Execution Successful ===
```